COMPUTER SCIENCE & IT

DIGITAL LOGIC

Lecture No: 04

Sequential Circuit

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Recap of Previous Lecture

Race Around condition

Counter

Counter

- $0-0-1-1-2-2-2-3-3-3-3 \longrightarrow$ MOD no. $= 11$

- $0-1-3-2 \longrightarrow$ MOD no. $= 4 \longrightarrow 2$ ff

$Q_1 \, Q_0$

$$00 \xrightarrow[1^{st}]{} 01 \xrightarrow[2^{nd}]{} 11 \xrightarrow[3^{rd}]{} 10 \xrightarrow[4^{th}]{} 00 \xrightarrow[5^{th}]{} (01)$$

- **Note:** - After application of MOD no. of clock pulses counter will again be at the starting state.
  - After application of integer multiple of MOD no. of clock pulses counter will again be at starting state.

if starting state is $(3)_{10}$ then after 263 clock pulses, what will be the state of the counter $\underline{(1)_{10}}$

$$(3)_{10} \xrightarrow{4\,clk} (3)_{10} \xrightarrow{4\,clk} (3)_{10}$$

$$(3)_{10} \xrightarrow[260]{65\times4} (3)_{10} \xrightarrow{261} 2 \xrightarrow{262} 0 \xrightarrow{263} 1$$

$f_{in} = f_{clk}$ → frequency of external clock

$f_{out} = f_{clk}/n$

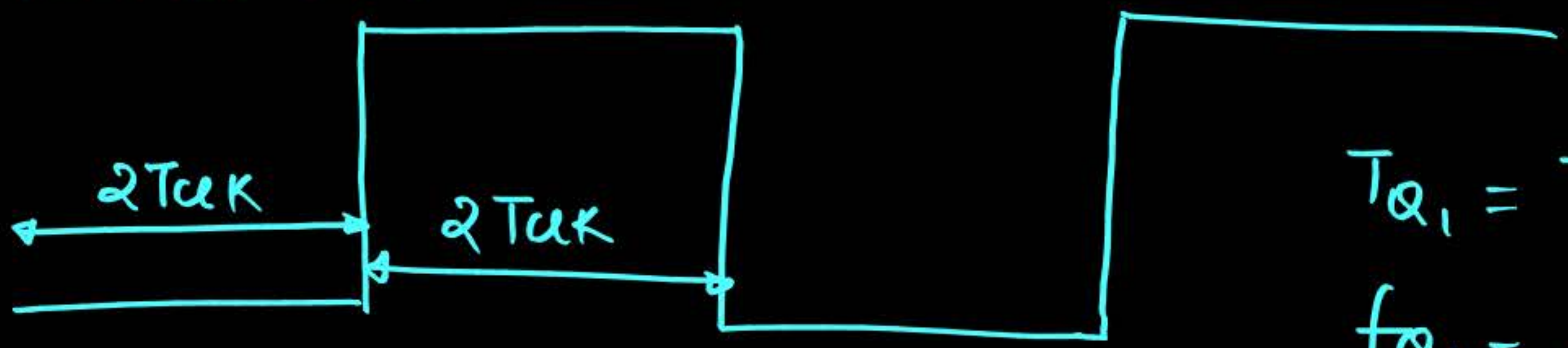
- 0 − 1 − 3 − 2

$f_{clk}/4$

Q_1	Q_0
0	0
0	1
1	1
1	0
0	0

MOD no. = 4

$Q_1 \rightarrow T_{ON} = 2T_{clk}$, $T_{off} = 2T_{clk}$



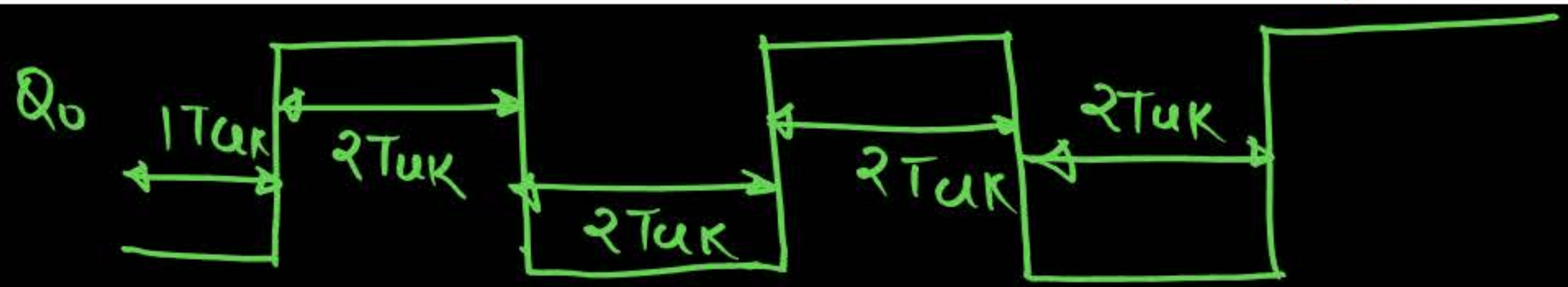
$T_{Q_1} = T_{ON} + T_{off} = 4T_{clk}$

$f_{Q_1} = \frac{1}{T_{Q_1}} = \frac{1}{4T_{clk}} = \frac{f_{clk}}{4}$

$Q_0 \rightarrow T_{ON} = 2T_{clk}$, $T_{off} = 2T_{clk}$

$T_{Q_0} = T_{ON} + T_{off} = 4T_{clk}$

$f_{Q_0} = f_{clk}/4$



- $0 - 1 - 4 - 3 - 2 - 5$ $\quad$ MOD no $= 6$

Q_2	Q_1	Q_0
0	0	0
0	0	1
1	0	0
0	1	1
0	1	0
1	0	1

$Q_0 \longrightarrow T_{off} = 1T_{clk}, \; T_{ON} = 1T_{clk}$

$T_{Q_0} = 2T_{clk} \quad f_{Q_0} = f_{clk}/2$

$Q_1 \longrightarrow T_{off} = 4T_{clk}, \; T_{ON} = 2T_{clk}$

$T_{Q_1} = 6T_{clk} \longrightarrow f_{Q_1} = f_{clk}/6$

$Q_2 \longrightarrow T_{off} = 2T_{clk}, \; T_{ON} = 1T_{clk}$

$T_{Q_2} = 3T_{clk}, \quad f_{Q_2} = f_{clk}/3$

- 0 — 1 — 4 — 2 MOD no = 4

$\hookrightarrow$ 3 ff

Q_2	Q_1	Q_0
0	0	0
0	0	1
1	0	0
0	1	0

$Q_0 \longrightarrow T_{ON} = 1T_{clk}$ $\quad T_{off} = 3T_{clk}$

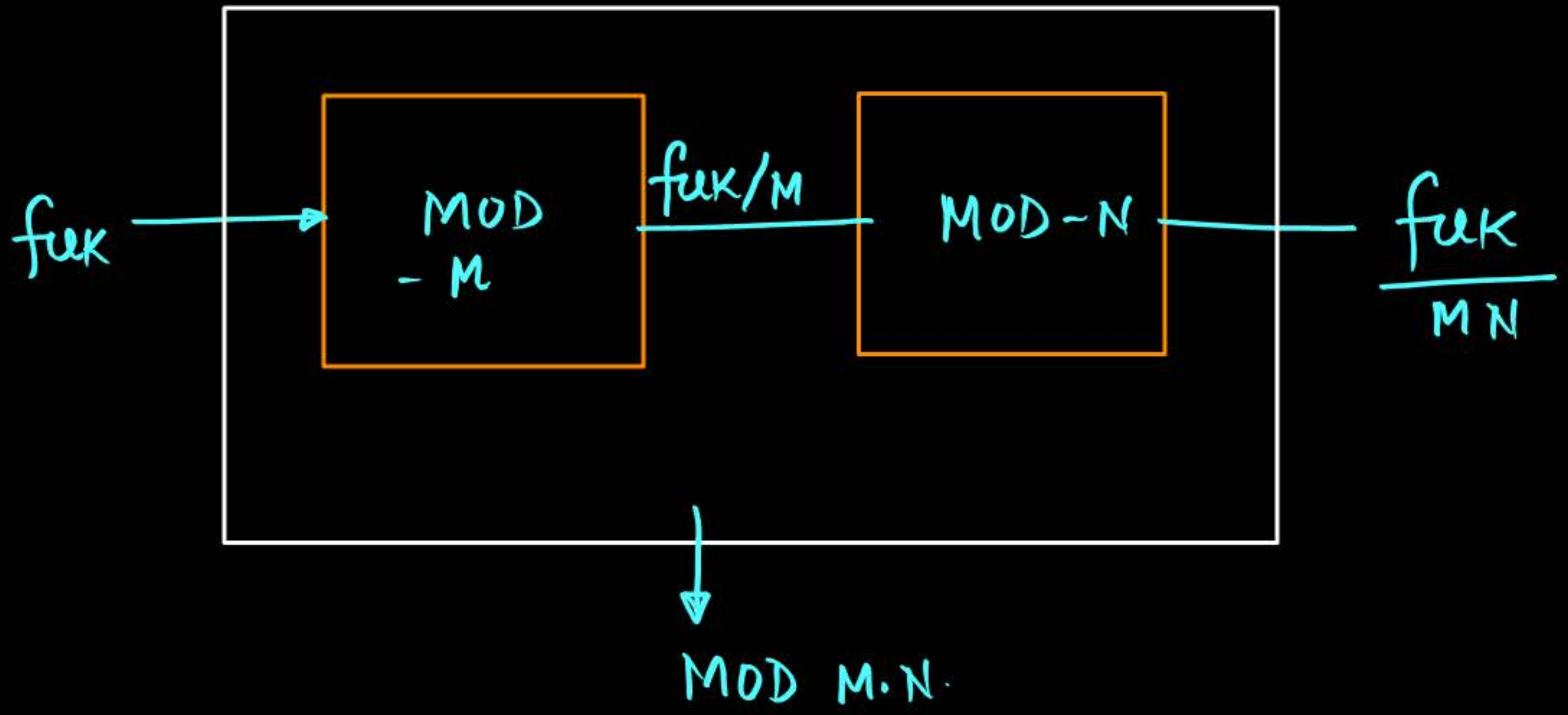
$T_{Q_0} = 4T_{clk}, \quad f_{Q_0} = \frac{1}{4} f_{clk}$

$Q_1 \longrightarrow T_{ON} = 1T_{clk}, \quad T_{off} = 3T_{clk}$

$T_{Q_1} = 4T_{clk} \quad f_{Q_1} = f_{clk}/4$

$Q_2 \longrightarrow T_{ON} = 1T_{clk}, \quad T_{off} = 3T_{clk}$

$T_{Q_2} = 4T_{clk}, \quad f_{Q_2} = f_{clk}/4$



Q. We have a MOD-4 counter and MOD-10 counter cascaded, then what is the resultant MOD no. $\underline{40}$.

Types of Counter

Asynchronous Counter

- Here different ffs are driven by different clocks.
- Only fix sequence is possible
  - up sequence
  - down sequence
- eg. up counting sequence
  down counting sequence
- It is relatively slower.

Synchronous Counter

- Here all the ffs are driven by same clock.
- All possible sequences can be designed.

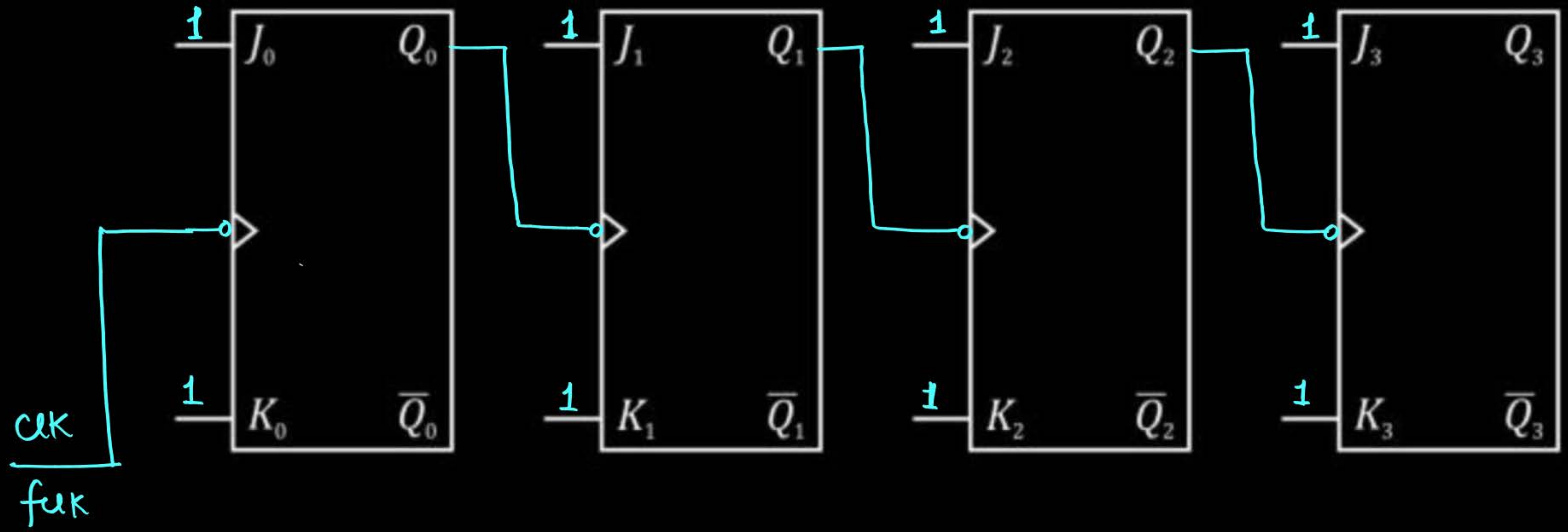
eg. Ring Counter, Johnson Counter etc
Any sequence.

- It is relatively faster.

4-bit Asynchronous Counter

condition for asynchronous ff:
1. only one direct clock
2. next ff should trigger by just previous ff
3. should run on toggle mode that is j = 1 and k = 1

- Circuit :



- Working :

- $Q_0 \longrightarrow$ toggles in every clock cycle.
- $Q_1 \longrightarrow$ toggles when Q_0 take transition from $(1-0)$.
- $Q_2 \longrightarrow$ toggles when Q_1 take transition from $(1-0)$
- $Q_3 \longrightarrow$ toggles when Q_2 take transition from $(1-0)$

	Q_3	Q_2	Q_1	Q_0
Starting state	0	0	0	0
1st clock	0	0	0	1
2nd clock	0	0	1	0
3rd clock	0	0	1	1
4th clock	0	1	0	0
5th clock	0	1	0	1
6th clock	0	1	1	0
7th clock	0	1	1	1
8th clock	1	0	0	0
9th clock	1	0	0	1

MOD-16 Counter.

$Q_0 \rightarrow T_{off} = 1T_{clk}, T_{ON} = 1T_{clk}$

$T_{Q_0} = 2T_{clk} \rightarrow f_{Q_0} = f_{clk}/2$

$Q_1 \rightarrow T_{off} = 2T_{clk}, T_{ON} = 2T_{clk}$

$T_{Q_1} = 4T_{clk}, f_{Q_1} = \frac{f_{clk}}{4}$

$Q_2 \rightarrow T_{off} = 4T_{clk}, T_{ON} = 4T_{clk}$

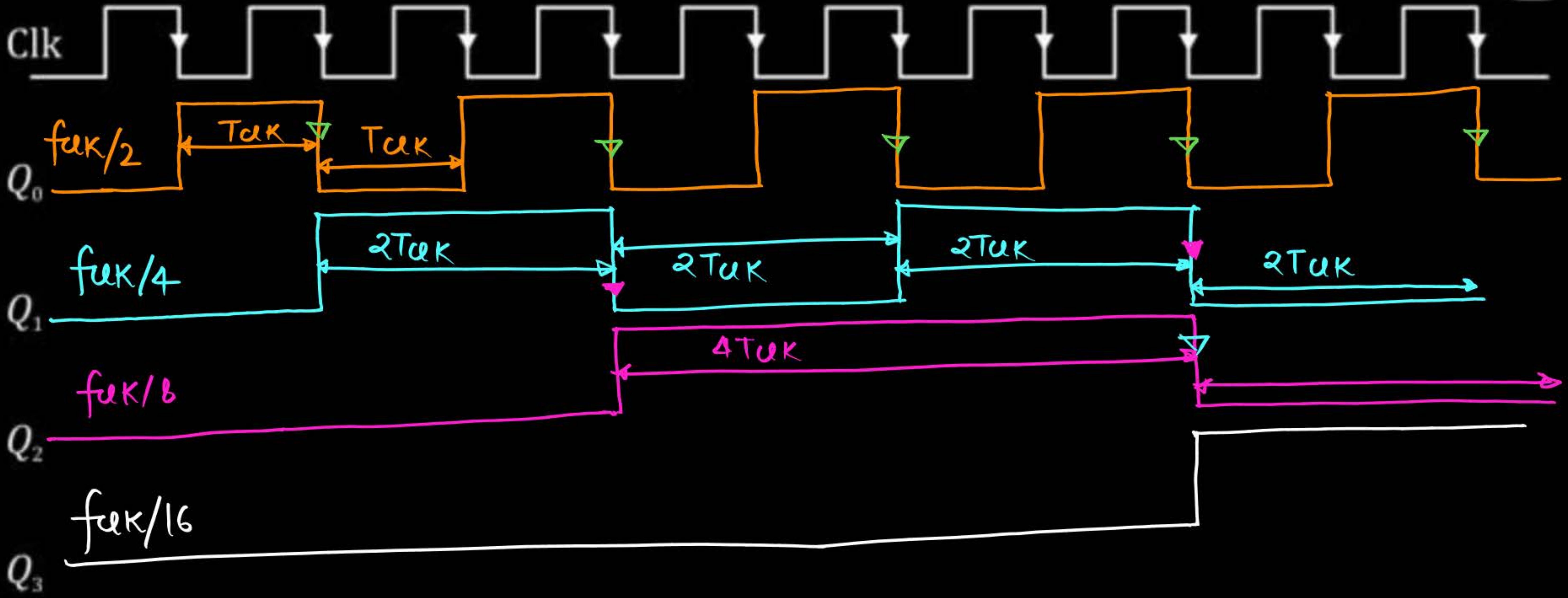
$T_{Q_2} = 8T_{clk}, f_{Q_2} = f_{clk}/8$

$Q_3 \rightarrow T_{off} = 8T_{clk}, T_{ON} = 8T_{clk}$

$T_{Q_3} = 16T_{clk}, f_{Q_3} = f_{clk}/16$

10^{th}	1 0 1 0
11^{th}	1 0 1 1
12^{th}	1 1 0 0
13^{th}	1 1 0 1
14^{th}	1 1 1 0
15^{th}	1 1 1 1
16^{th} clock	0 0 0 0

- Clock Diagram :



Topic : 2 Min Summary

→ Asynchronous Counter.

Thank you

GW Soldiers !